Analysis of Red-Fluorescent Proteins Provides Insight into Dark-State Conversion and Photodegradation

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ABSTRACT Fluorescent proteins (FPs) are powerful tools that permit real-time visualization of cellular processes. The utility of a given FP for a specific experiment depends strongly on its effective brightness and overall photostability. However, the brightness of FPs is limited by dark-state conversion (DSC) and irreversible photobleaching, which occur on different timescales. Here, we present in vivo ensemble assays for measuring DSC and irreversible photobleaching under continuous and pulsed illumination. An analysis of closely related red FPs reveals that DSC and irreversible photobleaching are not always connected by the same mechanistic pathway. DSC occurs out of the first-excited singlet state, and its magnitude depends predominantly on the kinetics for recovery out of the dark state. The experimental results can be replicated through kinetic simulations of a four-state model of the electronic states. The methodology presented here allows light-driven dynamics to be studied at the ensemble level over six orders of magnitude in time (microsecond to second timescales).

INTRODUCTION

Genetically encodable fluorescent proteins (FPs) are invaluable tools for illuminating biochemical processes within the dynamic cellular context. Requiring only molecular oxygen, the chromophore is formed autocatalytically within the interior of an 11-stranded β-barrel (1). Additionally, amino acid mutations proximal to the chromophore perturb the spectral attributes, permitting the generation of FPs that span the visible spectrum as well as variants with unique photoactivation and photoswitching characteristics (1,2).

Because of their unique fluorescent attributes, FPs have been the subject of intense theoretical and experimental analyses. For example, spectroscopic work on FPs has revealed an intricate chromophore environment involving excited-state proton transfer (3), a β-barrel-mediated dynamic Stokes shift (4), and a quadratic Stark effect (5). Despite this wealth of information, it remains poorly understood why FPs routinely emit 10–100$\times$ fewer photons than do small-molecule fluorescent dyes (6,7). One mechanism that may contribute to this diminished photon output is dark-state conversion (DSC, also known as reversible photobleaching or blinking). In green fluorescent protein (GFP) and its yellow-emitting variants, investigators have identified two DSC processes that occur on the 0.01–1 ms timescale (8–10). One, which is pH-dependent, is attributed to protonation of the p-hydroxybenzylidene moiety, and the other, which is pH-independent, is attributed to a conformational change of the chromophore and/or its environment into a nonradiative configuration. Additionally, a separate submicrosecond process has been identified in GFP, likely involving intersystem crossing to the triplet state, that when given sufficient time for relaxation, increases total photon output before photobleaching >20-fold (11).

In red-emitting FPs (RFPs), DSC can be both pH-sensitive (12,13) and pH-insensitive (8,14,15), with photophysical processes taking place on the μs (e.g., the triplet state) and 0.1 ms (conformational dynamics) timescales (16). Although investigators have examined a variety of RFPs, the published models are in disagreement and provide little insight into irreversible photobleaching. As a result, it is not clear whether irreversible photobleaching occurs out of these transient and long-lived dark states. In this study, we analyze multiple closely related proteins to explore how diverse photophysical properties coevolve with one another. Our goal is to provide a model that combines DSC and irreversible photobleaching in the context of additional photophysical properties (e.g., quantum yield and extinction coefficient), and to shed light on how these properties change upon mutation of the protein structure.

TagRFP and a closely related variant, mKate, are ideal candidates for evaluating how photophysical properties vary within a series of FPs (17). mKate was first derived from TagRFP by directed evolution with selection pressure for red-shifted emission, and characterized by the incorporation of four mutations (R67K, N143S, F174L, and H197R) (17,18). Subsequent studies identified a single mutation in TagRFP, S158T, that improved its photostability ninefold and is referred to here as TagRFP-T (7). In mKate, an S158A mutation (hereafter referred to as mKate2) improved the brightness (defined as the product of the extinction coefficient and quantum yield) 2.8-fold (7,19). In light of the limited number of mutations that are needed to evaluate

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this pathway, as well as its diverse phenotypes and sensitivity to modest structural perturbations (e.g., S158T), we chose to characterize the photophysics of this system and introduce a pulsed photoexcitation method that separately resolves the magnitudes of irreversible photobleaching and DSC.

MATERIALS AND METHODS

FPs were analyzed in vitro (15 mM MOPS, 100 mM KCl, pH 7.0) to determine their fluorescence lifetime, excitation maxima, emission maxima, absorption spectra, and quantum yield. All photobleaching measurements were performed on freely diffusing nuclear localized FPs within living adherent HeLa cells. Kinetic analysis and numerical modeling were performed within a commercial software package. Details pertaining to the in vitro FP characterization, mammalian cell culture, in vivo photobleaching measurements, photobleaching data analysis, and kinetic simulations are provided in the Supporting Material.

RESULTS

Spectral changes associated with mutations

To characterize the photophysical properties of a series of closely related FPs, we generated variants of TagRFP with combinations of the four mutations that convert TagRFP into mKate (R67K, N143S, F174L, and H197R). These mutations are illustrated in Fig. 1, which depicts the chromophore environment of TagRFP and mKate. We also incorporated additional mutations (S158A/C/T and H197I/Y) to explore the influence of these amino acid substitutions on photostability and red shift, respectively (20). A total of 27 proteins were generated, purified, and compared with six mFruits (mApple (7), mCherry (21), mOrange (21), mOrange2 (7), mStrawberry (21), and mPlum (22)). Table S1 lists the excitation and emission wavelength, extinction coefficient, quantum yield, and fluorescence lifetime measured for each protein. Not surprisingly, these parameters varied widely across the proteins, and the influence of each individual mutation on the photophysical properties was strongly dependent on context (i.e., on the other mutations present). For example, incorporation of F174L into TagRFP R67K S158T caused a dramatic reduction in the quantum yield from 0.36 to 0.04, but the same mutation introduced into TagRFP N143S S158T caused a slight increase in the quantum yield from 0.25 to 0.40.

It was previously shown that TagRFP and mKate crystal structures reveal *trans* and *cis* chromophore configurations, respectively (Fig. 1) (19,23), consistent with spectroscopic and electronic structure calculations that suggested *trans*-to-*cis* isomerization of the p-hydroxybenzylidene moiety into a negatively charged electrostatic environment would cause a red shift in the excitation (24,25). For the mutant proteins examined here, the maximum excitation wavelength generally clustered in two groups, perhaps corresponding to proteins with chromophores in either the *trans* or *cis* configuration (Fig. 1 *c*). However, the excitation spectra of some proteins exhibited a bimodal distribution (i.e., TagRFP N143S S158T; Fig. 1 *d*), suggesting that a mixture of two chromophore configurations exist throughout the transformation of TagRFP-T to mKate2. These results suggest that the change in free energy between the *cis* and *trans* ground-state configurations may be relatively small and acutely sensitive to the mutational context in TagRFP-T variants.

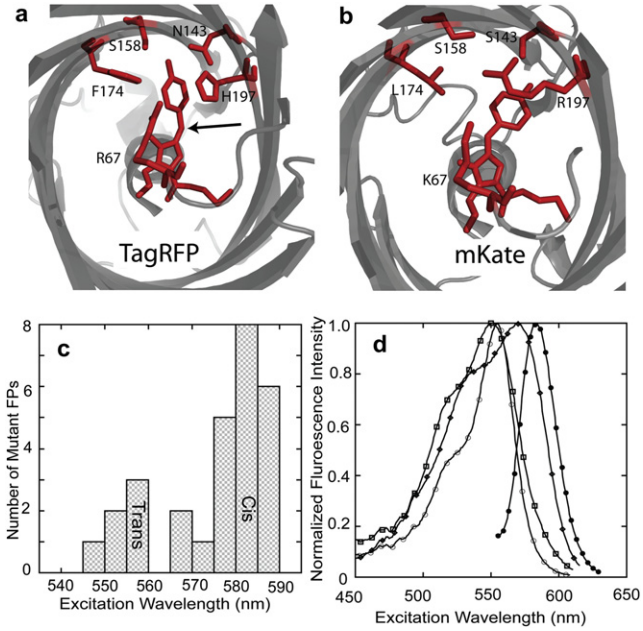

FIGURE 1 Structural and spectral changes associated with FPs. (*a* and *b*) Crystal structures of the chromophore pocket for TagRFP (*a*) and mKate (*b*). The crystal structures show TagRFP (PDB 3M22) and mKate (PDB 3BXB) in the *trans* and *cis* configurations, respectively, due to rotation around the bond marked by the arrow. Mutations explored in this study include R67K, N143S, S158A/C/T, F174L, and H197I/R/Y. (*c*) A bimodal distribution of excitation wavelengths, likely indicative of a mixture of the *trans* and *cis* configurations of the chromophore throughout the transformation of TagRFP to mKate. Table S1 summarizes the measured extinction coefficients, quantum yields, fluorescence lifetimes, and excitation and emission wavelengths for these mutant FPs. (*d*) Excitation spectra of single mutants in the TagRFP-T background: TagRFP-T (*open circle*), TagRFP-T R67K (*open square*), TagRFP-T N143S (*black diamond*), and TagRFP-T F174L (*black circle*). Note the broadened excitation spectra for TagRFP-T N143S, likely indicative of two ground-state configurations, both of which absorb.

Ensemble photobleaching: differentiating DSC and irreversible photobleaching

Photobleaching (i.e., the gradual decay of fluorescence upon exposure to light) significantly limits the photon output of FPs; however, the mechanisms of fluorescence decay remain poorly characterized. To examine photobleaching for a panel of FPs in vivo, we expressed freely diffusing, nuclear localized FPs in HeLa cells and continuously illuminated them using a Xe arc lamp or continuous-wave (CW) laser.

Fig. S1 provides a representative image of FP localization and the extent of photobleaching observed. We selected TagRFP-T mutants from the previous experiments to include well-maturing variants (i.e., predominantly red-absorbing) with diverse spectral properties, which allowed us to assess how photobleaching correlates with different photophysical attributes (e.g., quantum yield and fluorescence lifetime).

Fig. 2 shows that the FPs exhibit a wide range of photobleaching behaviors and kinetics upon exposure to either wide-field or laser (2.5 kW/cm^2 and 25 kW/cm^2) illumination. The observed responses include mono- and multiexponential decay, photoactivation, and rapid decreases in fluorescence intensity followed by a transient increase and subsequent decay. As expected, increasing the illumination intensity led to faster photobleaching. However, there were also unexpected responses, suggesting that FPs may exhibit different mechanisms of photobleaching upon wide-field versus laser illumination or at different intensities of laser illumination. For example, TagRFP-T undergoes photoactivation with wide-field illumination (Fig. 2 *a*). However, when it is illuminated at 25 kW/cm^2, the same protein undergoes a rapid decrease in fluorescence intensity followed by multiexponential decay with negligible photoactivation (Fig. 2 *c*). The inset in Fig. 2 *c* highlights the fact that at 25 kW/cm^2, photobleaching is characterized by a rapid decrease in fluorescence intensity during the first 5 ms of illumination, followed by a slower decay.

The most commonly observed behavior involved rapid decay followed by a slower decrease. Because fluorescence decay occurred over a wide range of timescales, decay curves were interpolated and converted to time points equally spaced over six orders of magnitude in log-time. Fig. 3 *a* shows a typical FP fluorescence decay curve in log time. Decay is characterized by three separate kinetic phases: 1), an initial monoexponential decay (<800 μs); 2), a steady-state phase (800 μs to 5 ms) during which fluorescence intensity remains constant; and 3), a gradual biexponential phase (>5 ms).

To gain insight into this complex behavior, we compared photobleaching upon continuous illumination with photobleaching using a train of 2 ms 25 kW/cm^2 pulses separated by 8 ms dark periods. Hereafter, we refer to this excitation scheme as pulsed excitation. Fig. 3 *b* shows a characteristic photobleaching curve using pulsed excitation. The inset of Fig. 3 *b* demonstrates that the initial monoexponential decay phase (i.e., <800 μs) observed during continuous illumination was replicated in each excitation pulse, and was largely reversible. Accordingly, we hypothesized that this rapid decay corresponds to conversion to a transient dark state. In this context, the term "dark state" refers to a state that is nonfluorescent, less fluorescent, less absorbing, or nonabsorbing at the wavelength used (e.g., the protonated chromophore, or the triplet state).

To quantify the extent of fluorescence recovery, we defined the percent recovery as $(F_R - F_B)/(F_L - F_B)$, where F_L and F_R are the initial fluorescence intensities of the first and second excitation pulses, respectively; and F_B is the final fluorescence intensity of the first excitation pulse (Fig. 3 *b*, *inset*) (7). The percent recovery values for each protein are listed in Table S2 and vary from 55% to 100%. Fluorescence recovery appeared to be complete within 8 ms, as prolonged durations in the dark (up to 10 s) did not lead to statistically significant increases in percent fluorescence recovery (TagRFP, mOrange2, and mCherry; analysis of variance, $P > 0.05$). However, in some cases (e.g., mKate2), the percent recovery changed depending on the number of pulse exposures, presumably due to residual dark-state accumulation.

To quantify the photobleaching of different FPs and to differentiate irreversible photobleaching from DSC, we fit

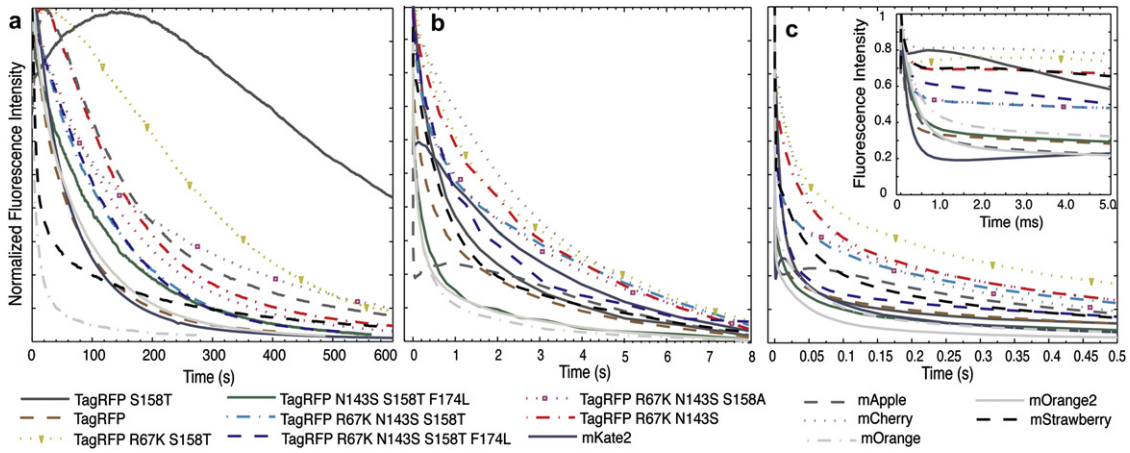

FIGURE 2 TagRFP-T mutant and mFruit photobleaching spectra. Continuous photobleaching curves were collected under (*a*) wide-field illumination conditions ($\approx$100 W/cm^2) and (*b*) 2.5 kW/cm^2 and (*c*) 25 kW/cm^2 532 nm laser illumination. The inset in *c* provides closer inspection of the first 5 ms for the photobleaching decay. Each line is representative of the photobleaching observed for a single cell. Details of acquisition and analysis are provided in the Supporting Material.

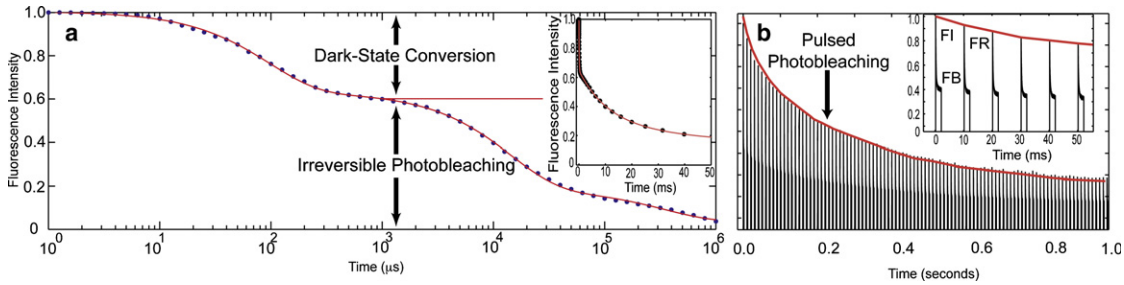

FIGURE 3 Photobleaching decay for TagRFP R67K N143S S158T F174L. (*a*) Photobleaching kinetics and the respective triexponential fit (*red*) in log time. Initial decay representative of DSC (i.e., <1 ms), steady-state plateau (i.e., ≈ 1 ms), and irreversible photobleaching phase (i.e., >5 ms). The inset in panel *a* shows photobleaching kinetics plotted in linear time. (*b*) Pulsed photobleaching kinetics (*red line*) is defined as the exponential decay of the peak fluorescence intensity of each pulse under pulsed excitation. The inset in *b* shows the percent recovery after 2 ms excitation and 8 ms in the dark. Percent fluorescence recovery is defined as (FR − FB)/(FL − FB), where FL and FR are defined as the fluorescence intensities of the initial and second pulses, respectively, and FB is the final fluorescence intensity after 2 ms of excitation.

the data to a sum of exponentials. The fitted rate constants enabled us to evaluate the time constants (defined as the reciprocal of the rate constant) for the different phases of fluorescence decay. At 25 kW/cm^2, continuous photobleaching data were fit to a sum of three exponential decays, allowing the kinetics of the fast and the weighted average of the slow biexponential phase to be independently determined. A representative fit is shown in Fig. 3 *a*. Because our pulsed excitation suggested that the initial fast decay was largely reversible, this phase is referred to as DSC. To evaluate the effect of the cellular environment on the observed photophysics, we measured the kinetics and percent DSC for a representative FP (mCherry) in vitro. Relative to our in vivo data, DSC increased 7% in vitro, presumably due to environmental factors. Furthermore, we compared the kinetics and percent DSC with published results from fluorescence correlation spectroscopy (FCS) on the same protein (13), and further discrepancies were attributed to difficulties in differentiating reversible and irreversible fluorescence fluctuations in FCS (for a detailed discussion, see "mCherry Analysis" in the Supporting Material). Conversely, the second slower phase appeared to be irreversible and hence is referred to as irreversible photobleaching. Table S2 summarizes the parameters obtained from the fits of 14 different proteins, including the amplitude of DSC (defined as the percentage of the total decay attributable to DSC), as well as the time constants for DSC and irreversible photobleaching for FPs exposed to 25 kW/cm^2. Besides a small decrease in fluorescence intensity (≈1–4%) at 10 μs, no convincing evidence of triplet state dynamics was observed. At 2.5 kW/cm^2, the steady-state plateau was less pronounced, and thus the more gradual part of the photobleaching was fit to a biexponential decay (weighted time constant presented in Table S2). Under wide-field conditions, the three phases were not broadly identifiable, and consequently this approach was not used to fit these data.

To quantitatively assess whether photobleaching occurs out of transient dark states, we also performed photobleaching assays under pulsed excitation conditions. For each FP, we determined the time constant of irreversible photobleaching for pulsed excitation conditions by locating the maximum fluorescence intensity (i.e., F_R) for each excitation pulse and fitting the decrease in peak fluorescence intensity to a monoexponential decay (*inset* and *red curve* in Fig. 3 *b*). Hereafter, the results from this kinetic analysis are referred to as the pulsed irreversible photobleaching time constant (values are presented in Table S2). For all of the FPs studied, photobleaching under pulsed excitation is slower than photobleaching under continuous illumination. The extreme cases are mCherry, which exhibits a 13-fold gain in the photobleaching time constant under pulsed illumination, and TagRFP R67K N143S S158T, which shows only a twofold gain. This result suggests that for some FPs, irreversible photobleaching from dark states is minimized by pulsed excitation when the pulse separation is sufficient for these states to depopulate between excitation pulses.

Comparison of irreversible photobleaching in FP variants

By measuring photobleaching in a panel of FPs under different illumination conditions, we were able to identify general trends and hence common themes in fluorescence decay. Fig. 4 *a* compares the irreversible photobleaching time constant under continuous illumination at 2.5 and 25 kW/cm^2. Overall, time constants at 2.5 kW/cm^2 were significantly greater than at 25 kW/cm^2, indicating a slower rate of fluorescence decay at lower-intensity illumination. In general, the time constants were correlated so that FPs that were more susceptible to photobleaching at 2.5 kW/cm^2 were also more susceptible at 25 kW/cm^2 (see mOrange and mOrange2). However, some FPs showed heightened sensitivity to increases in excitation intensity. For example, mCherry was 30% less photostable than TagRFP R67K S158T at 25 kW/cm^2, but twofold more photostable at 2.5 kW/cm^2. This observation points to the need to

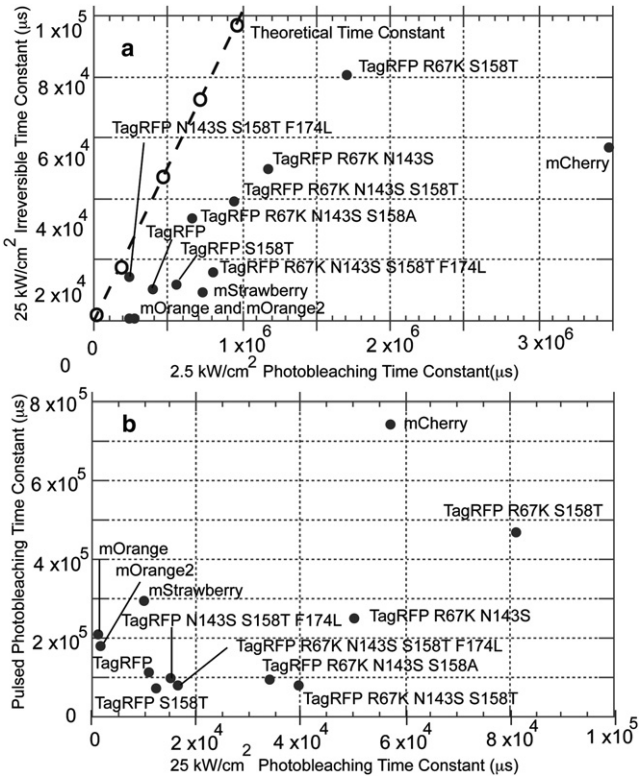

FIGURE 4 Comparison of irreversible photobleaching parameters of the different FPs. (*a*) Correlation between irreversible photobleaching time constants at 2.5 kW/cm^2 and 25 kW/cm^2 laser illumination shows that different proteins have different sensitivities to heightened excitation rates, and the rank order of photostability changes with intensity. The dashed line shows the anticipated correlation for a simple three-state system (e.g., ground, excited, and bleached) in which 10-fold increases in excitation intensity result in 10-fold decreases in the photobleaching time constant. (*b*) Comparison of the irreversible photobleaching time constant obtained for continuous versus pulsed illumination. Under pulsed illumination, FPs with photoreactive dark states become more photostable, whereas FPs with photoprotective dark states do not.

understand photostability in terms of photoexcitation rates relative to timescales of excited-state population transfer. Fig. 4 *b* compares the photobleaching time constant for pulsed versus continuous illumination at 25 kW/cm^2. Here, the two parameters are poorly correlated among the proteins tested, suggesting that some proteins experience gains in photostability when subjected to pulsed excitation and others do not. For example, although mCherry is less photostable than TagRFP R67K S158T under continuous illumination at 25 kW/cm^2, it becomes more photostable when subjected to pulsed excitation at the same intensity.

Comparison of DSC in FP variants

Fig. 5 *a* presents a comparison of the percent DSC (i.e., the fast reversible phase of photobleaching under continuous illumination) and the irreversible photobleaching time constant (i.e., the slow and irreversible phase under continuous illumination) at 25 kW/cm^2. For these FPs, as the percent DSC increases, the photobleaching time constant decreases (correlation indicated by the *dashed line*), suggesting that as DSC increases, the propensity to photobleach also tends to increase (as observed for mOrange, mOrange2, TagRFP, etc.). However, there are significant exceptions. For example, a single mutation in TagRFP S158T solely affects DSC (Fig. 5 *a*, *arrow 1*), whereas incorporation of R67K into TagRFP S158T exclusively modulates irreversible photobleaching (Fig. 5 *a*, *arrow 2*). Conversely, some mutations simultaneously modulate both DSC and irreversible photobleaching rates (Fig. 5 *a*, *arrow 3*).

To examine the effect of excited-state lifetime on DSC, we compared the fluorescence lifetimes of purified proteins (Table S1) with the observed DSC kinetics for FPs in cells. Fig. 5 *b* reveals a correlation between increasing fluorescence lifetime and increased percent DSC. At 25 kW/cm^2, depending on the absorption cross-section, on average one photon is absorbed every 40–80 ns (see Supporting Material), suggesting that transient absorption out of the first excited singlet state is unlikely. Furthermore, the radiative rate, as estimated from the fluorescence quantum yield and lifetime ($\phi_{fl} = k_{rad} \times \tau_{fl}$), was found to be independent of the mutational context. These results suggest that DSC is competitive with radiative decay from the first excited singlet state. Consequently, FPs that exhibited the longest fluorescence lifetimes, and hence had the greatest percent DSC and propensity to undergo irreversible photobleaching, also tended to have the largest quantum yields (see "Relationship between quantum yield and DSC" in the Supporting Material for a detailed discussion).

As mentioned above, both the amplitude and kinetics of DSC varied substantially for different FP variants. Fig. 5 *c* shows the time constant of DSC versus percent DSC for different FPs. Of interest, this comparison reveals that RFPs that have a slower rate of DSC, and hence reach the steady-state phase of photobleaching more slowly, have a greater percent DSC. This observation will be explained by the kinetic modeling described below.

Kinetic modeling

A careful examination of the RFP photobleaching behavior at 25 kW/cm^2 revealed clear trends in the percent DSC, the measured rate of DSC, the fluorescence lifetime, and the measured rate of irreversible photobleaching. Given these observations, we sought to expand upon existing models for DSC to see whether we could quantitatively replicate the trends observed and, if so, gain additional insight into the mechanisms of DSC and irreversible photobleaching. Previously, Dickson et al. (26) proposed a four-state model consisting of two anionic and two neutral chromophore states to describe the blinking of yellow-emitting GFP variants at the single-molecule level. Here, we performed numerical simulations on an analogous four-state system

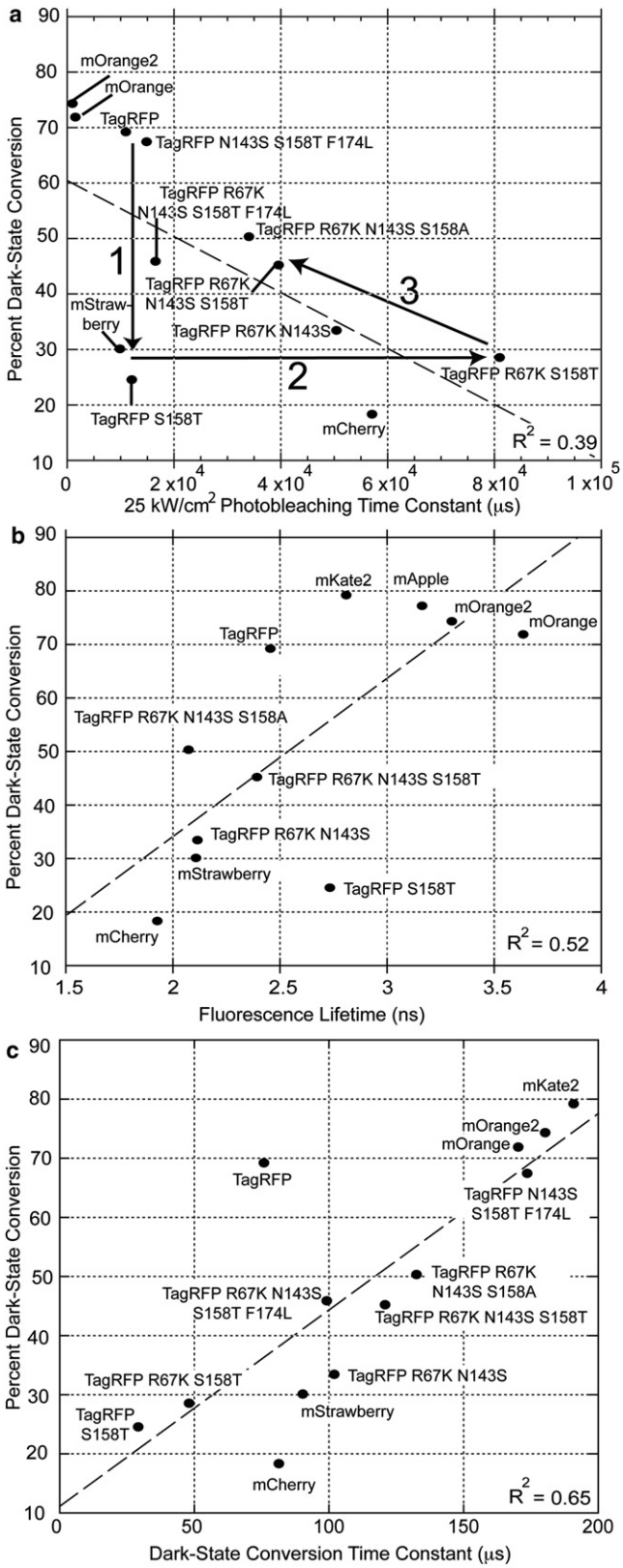

FIGURE 5 DSC correlation plots. (*a*) Single mutations perturb the percent DSC (*arrow 1*), irreversible photobleaching time constant (*arrow 2*), or both parameters simultaneously (*arrow 3*). A weak correlation (*dashed line*) suggests that increases in the percent DSC are accompanied by decreased photostability. (*b*) Comparison of the percent DSC and fluorescence lifetime suggests that DSC is competitive with emission from the first excited singlet state. (*c*) Percent DSC versus the time constant of DSC. Linear correlation reveals that the percent DSC increases in proteins with slower rates, or larger time constants, of DSC.

consisting of two bright states (S_0 and S_1) and two dark (or less-fluorescent) states (D_0 and D_1). In the context of RFPs, dark states likely represent a mixture of neutral (nonabsorbing at 532 nm) and/or isomerized (absorbing at 532 nm) chromophore states. A schematic of this model is presented in Fig. 6 *a*. The rates input into the kinetic simulations are referred to as microscopic, and the rates measured by fitting the results from the numerical simulations are referred to as the simulated time constant. The kinetic analysis from experimental data is referred to as measured.

In this four-state model, absorption of a photon (i.e., electronic transition from S_0 to S_1) is followed by depopulation through emission of a photon (k_{em}), nonradiative internal conversion, or a low-quantum-efficiency ($\phi_{dsc} = 1 \times 10^{-3}$, $k_{S1D1} \approx 5 \times 10^5\ s^{-1}$) (14) conversion to the weakly or nonradiative D_1 state. D_1 can decay to D_0 (k_{ic}), which can subsequently be converted back to S_0 (k_{dsr}). In accordance with Dickson et al. (26), recovery from D_0 to S_0 was assumed to be rate-limiting. As shown in Fig. 6 *b*, variation of the microscopic time constant for dark-state recovery (τ_{dsr}) from 1 to 500 μs altered the populations of the S_0 ground state and D_0 dark state (i.e., the percent DSC). This result suggests that the measured variation in the percent DSC (from 18% to 87%; Table S2) reflects changes in the kinetics of recovery from the dark state (τ_{dsr}). To test whether the DSC time constant obtained by fitting the fast phase of fluorescence decay reflects the rate of ground-state recovery (i.e., the transition from D_0 to S_0), we varied the microscopic time constant for dark-state recovery (τ_{dsr}) and determined the corresponding time constant for DSC from the numerical simulations. At fast timescales (i.e., <100 μs), the time constant of DSC is correlated with the microscopic dark-state recovery kinetics (Fig. 6 *c*), whereas at longer timescales (i.e., >100 μs) the parameters are uncorrelated. For some of the FPs (seven out of 13), the measured DSC time constant is within the linear range of the simulated parameters, suggesting that for these FPs the observed fluorescence decay directly reflects the kinetics of ground-state recovery.

The four-state model also explains the correlation between the fastest timescale of fluorescence decay and the magnitude of DSC. Here, FPs with slower rates of ground-state recovery will have an increased population buildup in D_0. In agreement with our experimental results, simulations predict that the percent DSC is linearly proportional to the DSC time constant (Fig. 6 *d*). Additionally, in agreement with our experimental data (Fig. 5 *b*), modeling confirms that DSC increases linearly with the lifetime of S_1 ($\tau_{fl} = 0.5$–4.0 ns), with small changes ($\approx$2-fold) in the simulated kinetics of DSC (results not shown).

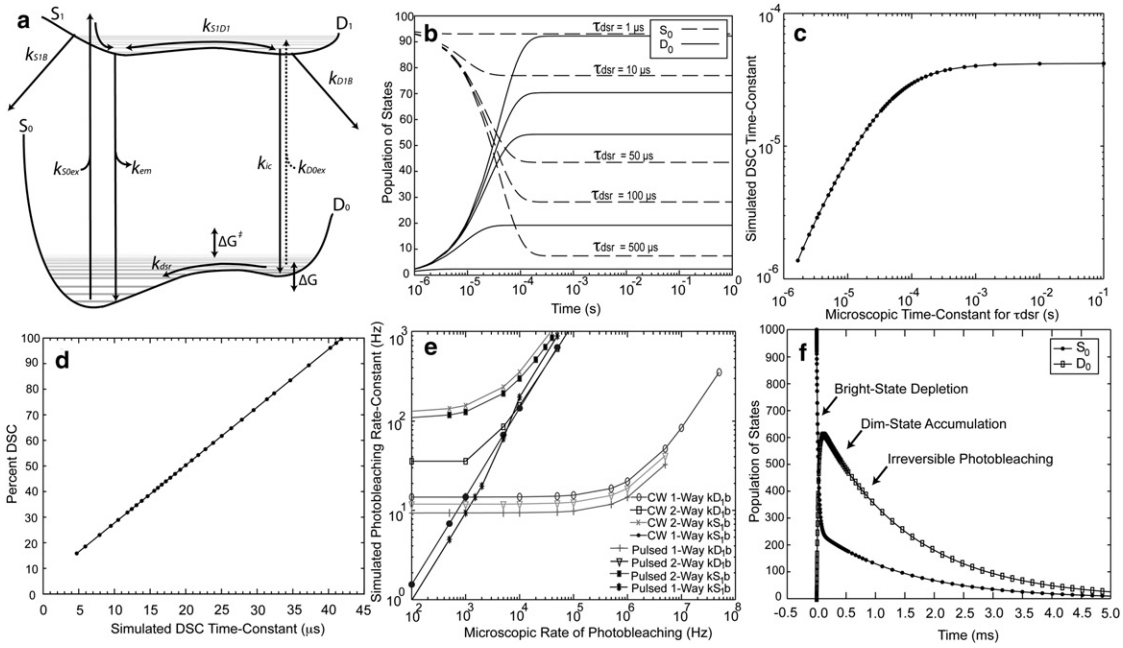

FIGURE 6 Numerical simulations of the proposed four-state model. (*a*) The four-state model includes the rate of excitation (k_{S0ex}), fluorescence emission (k_{em}), conversion from S_1 to D_1 (k_{S1D1}), internal conversion from D_1 to D_0 (k_{ic}), and dark-state recovery (k_{dsr}). Photobleaching was incorporated out of both S_1 (k_{S1B}) and D_1 (k_{D1B}). In some FPs, dark-state excitation (k_{D0ex}) is included and D_1 may be weakly fluorescent. (*b*) The influence of dark-state recovery kinetics on S_0 depletion at 25 kW/cm^2. As the microscopic time constant for dark-state recovery (τ_{dsr}) increases, the S_0 state is depleted and a significant increase in the D_0 population is observed. (*c*) Microscopic time constant of dark-state recovery versus the time constant obtained by fitting the simulated DSC. At fast timescales, the measured time constant of DSC accurately reflects the time constant of ground-state recovery, with increasing deviations observed for FPs with particularly slow DSC kinetics (long time constants). (*d*) Percent DSC versus time constant of DSC. The kinetics of DSC were determined by fitting the results from the numerical simulation. The model predicts that the percent DSC linearly increases with the time constant of DSC. (*e*) Numerical modeling of photobleaching out of S_1 and D_1. Kinetic modeling was performed in the presence (2-Way) and absence (1-Way) of dark-state excitation for both CW and pulsed illumination. The *x* axis represents the microscopic photobleaching rate for S_1 or D_1, and the *y* axis represents the photobleaching kinetics obtained by fitting the numerical simulations. For example, CW 1-Way D_1 Bleach represents a continuously illuminated numerical simulation in which the dark state does not absorb (i.e., one-way) and the D_1 bleaching rate is iteratively adjusted while the S_1 bleaching rate is held constant. Modeling demonstrates that when D_0 does not absorb, the observed photobleaching rate is independent of bleaching out of D_1 and linearly correlated with bleaching out of S_1. Under two-way conditions, D_1 bleaching becomes significant for continuous illumination but is minimized upon pulsed illumination. (*f*) The four-state model explains the complex photophysical behavior observed for mKate2 and mApple under continuous photobleaching at 25 kW/cm^2. In cases where D_1 may be weakly fluorescent due to changes in the fluorescence quantum yield or excitation rate, the rapid decrease and subsequent transient increase in fluorescence intensity represent population transfer from a bright state to a dim state before the onset of irreversible photobleaching.

Kinetic pathways for photodegradation out of both S_1 and D_1 (k_{S1B} and k_{D1B}, respectively; Fig. 6 *a*) were incorporated into the four-state model. Simulations of both CW and pulsed illumination experiments were performed as a function of both microscopic photobleaching rates (Fig. 6 *e*). In cases where excitation from D_0 to D_1 did not occur (1-Way in Fig. 6 *e*), the microscopic rate k_{S1B} was found to correlate linearly with the observed photobleaching rate for both pulsed and CW illumination. In this case, photobleaching from the dark state was negligible, and required k_{D1B} rates 1000-fold greater than k_{S1B} to significantly alter the observed photobleaching kinetics. This result indicates that nonabsorbing or weakly absorbing dark states are photoprotective. In the case where the excitation from D_0 to D_1 did occur (2-Way in Fig. 6 *e*), large differences were observed in the photobleaching rate between CW and pulsed illumination. For CW illumination, the photobleaching rate was no longer linearly correlated with changes in k_{S1B}, and changes in the observed photostability became significant for k_{D1B} at rates comparable to k_{S1B} (i.e., $>10^3$ s^{-1}). However, pulsed illumination minimized the contribution of k_{D1B} to the observed photobleaching kinetics. These simulations suggest that a comparison of photobleaching under pulsed and CW illumination provides insight into whether the dark state is photoprotective (bleaching does not occur from D_1) or photoreactive (bleaching does occur from D_1). For example, FPs with photoprotective dark states (e.g., TagRFP R67K S158T) likely do not absorb at the given excitation wavelength (i.e., the transition from D_0 to D_1 is insignificant).

DISCUSSION

Rapid, irreversible photobleaching and DSC remain major obstacles that limit the use of FPs for single-molecule applications, low-copy gene expression, and particle tracking

in vivo. A better understanding of DSC and new methods for measuring it will permit a detailed characterization of different FPs and may provide insight into which FP is most suited for a particular application. For example, our results (Fig. 2 *a*) and those of Shaner et al. (7) suggest that TagRFP-T may be the most photostable FP for low-excitation-intensity imaging (e.g., wide-field arc-lamp, total internal reflection fluorescence, and live-cell laser scanning confocal microscopy). However, in single-molecule applications (e.g., single-particle tracking and FCS), where excitation intensities exceed 1 kW/cm^2, mCherry and TagRFP R67K S158T appear to be substantially better than TagRFP-T (Fig. 2, *b* and *c*). These observations, as well as those regarding excitation wavelength dependence (7), illustrate that extensive data on excitation intensity and wavelength dependence data for each FP are necessary to select the optimum FP for a particular imaging application.

We explored whether a simple four-state model could explain the complex and often highly variable photophysical behavior of a panel of FPs. Numerical simulations demonstrate that a simple four-state model can explain how the magnitude of DSC varies with the kinetics of DSC and fluorescence lifetime, and how DSC contributes to irreversible photobleaching. The modeling also suggests that the measured kinetics of the initial decrease in fluorescence (i.e., kinetics of DSC) reflect the rate of dark-state recovery (τ_{dsr}), that the percent DSC is sensitive to changes in τ_{dsr}, and that a comparison of CW and pulsed excitation measurements provides insight into coupling of the dark state to photodegradation. In addition to replicating general trends, the four-state model can also explain the complex behavior of select FPs, such as mKate2 and mApple. These FPs undergo a rapid decrease in fluorescence intensity followed by a transient rise in fluorescence intensity before irreversible photobleaching occurs (Fig. 2, *b* and *c*). This behavior can be explained by considering population transfer from the initially excited bright state to a less fluorescent state (i.e., decreased quantum yield ϕ_{D1Fl} and/or absorption rate k_{D0ex}) rather than a strictly dark state. Consequently, a transient rise in fluorescence intensity represents a population buildup in the less-fluorescent state before the onset of irreversible photobleaching (Fig. 6 *f*).

The kinetic analysis of photobleaching at 25 kW/cm^2 revealed a clear trend in mutations that act synergistically or antagonistically to impact irreversible photobleaching. For example, the introduction of R67K into TagRFP S158T resulted in a 6.4-fold increase in the irreversible photobleaching time constant without significantly altering the percent DSC (Table S2). Likewise, single mutations in the context of TagRFP R67K S158T, including N143S, T158S, and T158A, all exhibited excellent photostability, although to a lesser extent than TagRFP R67K S158T alone. Of interest, in all tested cases, the presence of F174L decreased the photostability (Table S2), rendering the FP similar or worse in performance to the parent TagRFP S158T. These results suggest that mutations act in concert with regard to irreversible photostability, and that significant gains in photostability may be possible within the correct mutational context.

The kinetic simulations suggest that the rate of D_0 excitation dictates the extent to which D_1 is populated, and hence whether D_1 is a significant precursor along a photodegradative pathway. For example, at 25 kW/cm^2, mCherry is markedly more photostable under pulsed illumination than under CW illumination, suggesting that mCherry's dark state is photoreactive. Alternatively, TagRFP R67K S158T does not experience as large of a gain in photostability under pulsed conditions, suggesting that its dark state is more photoprotective than photoreactive. For TagRFP R67K S158T, our numerical modeling simulations suggest that this behavior could result from a decreased dark-state excitation rate, which could result from transient changes in the conformation or protonation state of the chromophore, whose kinetics are sensitive to the local environment.

The variability of photobleaching out of dark states also provides some explanation for the contrasting behavior of mCherry and TagRFP R67K S158T at 25 kW/cm^2 compared with 2.5 kW/cm^2. At moderate intensities (tens of kW/cm^2), the DSC rates scale linearly with excitation intensity (8). Consequently, FPs with more photoreactive dark states will show a heightened sensitivity to increases in excitation intensity relative to those with less reactive dark states. Our experimental results and kinetic simulations also provide some context for published results that demonstrate the excitation wavelength dependence of FP DSC and photostability (7,14,15). This wavelength dependence may be explained by changes in the excitation rate of D_0 relative to S_0, which varies in accordance with their corresponding excited-state absorption spectra. In cases where dark- to bright-state conversion occurs, the final steady-state distribution of dark state will depend on the rate of k_{D0ex}. For irreversible photobleaching, the relative rates of k_{D0ex} and k_{S0ex} dictate whether D_1 or S_1 will be the dominant pathway for photobleaching.

Our pulsed excitation method is advantageous for FPs primarily because DSC and photobleaching occur on widely varying timescales, which are probed by the microsecond time-resolved fluorescence transients that are repetitively observed during the millisecond excitation/dark intervals. The method therefore resolves kinetics over six orders of magnitude in time, and in particular extends measurements beyond the millisecond timescale. Experiments with a narrower experimental time window, such as FCS, do not directly resolve both timescales and therefore require a rigorous analysis of irreversible photobleaching (which has only been performed in a few cases (27–30)) for the DSC time constants to be accurately determined. Another key advantage of the broad time window in our method is that it can be employed over the three orders of magnitude in excitation intensities encompassed by many commonly

used imaging techniques, and thus potentially can provide one set of measurements for quantitative comparisons of signal intensities.

In conclusion, our examination of several closely related RFPs with pulsed excitation provides evidence for dark states of varying reactivity and highlights the role of these states in irreversible photobleaching. This work introduces a spectroscopic method for independently measuring DSC and irreversible photobleaching at the ensemble level. In comparison with DSC kinetics measured on single molecules (8,9,12–14), our methodology provides additional insight into slow events, such as irreversible photobleaching, without requiring surface immobilization, and allows measurements to be obtained inside living mammalian cells.

SUPPORTING MATERIAL

Additional Materials and Methods, a figure, two tables, and references are available at http://www.biophysj.org/biophysj/supplemental/S0006-3495(11)00786-7.

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